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RESEARCH ARTICLE

High-Speed CMOS Synchronous Binary Counter With Constant Counting Rate

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ABSTRACT In VLSI design, the counter stands as one of the most fundamental and widely utilized components. It finds extensive applications across various fields, including analog-to-digital converters, time-to-digital converters, frequency dividers, phase-locked loops, frequency synthesizers, and IoT networks. As technology advances, many applications necessitate high-speed counters supporting wider bit-widths. Nevertheless, the conventional synchronous binary counter, renowned for its simplicity, encounters a decline in counting rate as the counter size increases, primarily due to the lengthy signal chain. To overcome this limitation, a pre-scaled structure has been presented, ensuring a consistent counting rate even with increased sizes. In our pursuit of creating a high-speed counter, we incorporate this pre-scaled structure and introduce an early state detection technique to reduce the critical path affecting the counting rate. This enhancement expedites the generation of the enable signal for the subsequent sub-counter. Furthermore, to minimize the delay caused by T-flip-flops in the conventional binary counter, the sub-counters are intelligently designed based on the Johnson counter. In a 28nm CMOS process, the proposed counter achieves an impressive counting rate of 3.7 GHz, and maintains a nearly constant counting rate regardless of counter sizes, showcasing its robust performance.

INDEX TERMS Binary counter, constant time counter, high-speed counter, pre-scaled counter.

I. INTRODUCTION

Counters play a critical role in various applications, such as analog-to-digital converters (ADCs), time-to-digital converters (TDCs), frequency dividers, phase-locked loops (PLLs), frequency synthesizers, and IoT networks [1], [2], [3], [4], [5], [6]. As these applications continue to advance, there is an increasing demand for larger and faster counters. However, traditional binary counters face a trade-off between counter size and counting rate due to the presence of long signal chains [7], [8], resulting in low speed for large-sized counters. To overcome this challenge, extensive research has been

conducted to develop counters that maintain a consistent counting rate regardless of their size.

The asynchronous counter, also known as a ripple counter, operates without a universal clock. The main clock is connected only to the first flip-flop, while each subsequent flip-flop uses the output of the previous flip-flop as its clock signal. This structure offers some advantages in terms of energy efficiency and hardware complexity. However, it suffers from significant propagation delay, making it unsuitable for high-speed counting applications. As a result, recent research focuses on synchronous counters.

One approach involves the use of state generators. Various methods, such as pipelined chains of signal propagation based on systolic structures [9], [10], and linear-feedback shift

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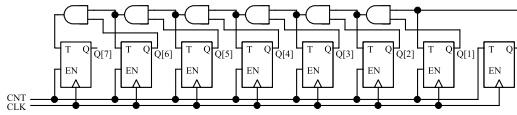


FIGURE 1. Conventional 8-bit binary counter.

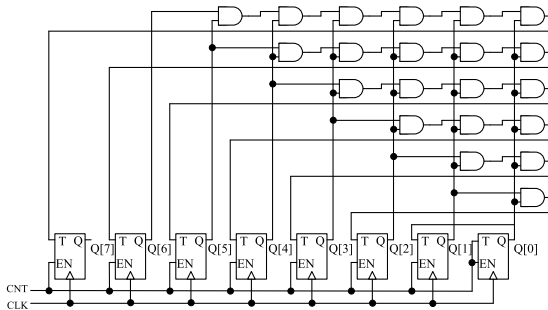


FIGURE 2. 8-bit binary counter with backward signal propagation.

registers (LFSRs) [11], [12], [13] have been presented to achieve a consistent counting rate irrespective of the counter size. Nevertheless, these methods have their limitations. Pipelined signal propagation chains lead to a doubling of flip-flop count, significantly increasing hardware complexity. On the other hand, the LFSR requires additional circuits for converting to binary states, and the number of states increases exponentially by increasing the LFSR size.

Another approach to maintain a consistent counting rate includes backward signal propagation [14] and pre-scaled counters that divide the counter into multiple sub-blocks [15], [16]. The backward signal propagation uses the property that the more significant bits become stable earlier than the less significant bits, eliminating the ripple propagation of signals. The overall speed is determined by the least significant bit (LSB), but the LSB has to go through numerous AND gates driving all the flip-flops. As the counter size increases, the large fan-out issue of the LSB becomes prominent, hindering the achievement of a truly consistent counting rate. Pre-scaled counters divide the counter into sub-blocks, and a pre-scaled enable (PEN) signal is generated from a low-order block. The PEN operates as an enable signal in the high-order block, and the lowest-order counter determines the overall counting rate. However, generating the PEN signal by using ring counters leads to a significant increase in hardware complexity [16]. In addition, the PEN signal has to drive many write-enable inputs in the next block, suffering from the large fan-out.

To leverage the advantages of the backward signal propagation and the pre-scaled structure while mitigating their drawbacks, a recent study [17] presented a novel counter design. This design utilized multiple 1-bit Johnson counters to generate as many PEN signals as the number of 1-bit Johnson counters, thereby reducing the hardware complexity of ring counters and the large fan-out problem of the PEN signal, increasing the counting rate significantly compared to the previous counters. However, the counter is designed based

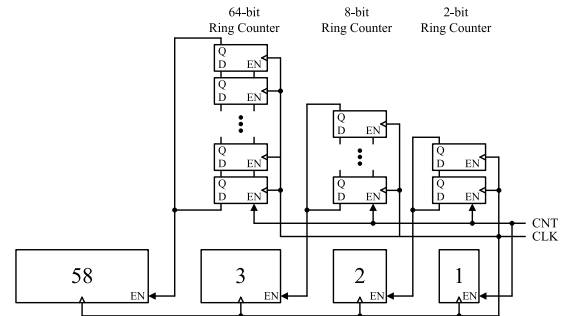


FIGURE 3. 64-bit pre-scaled counter using ring counters.

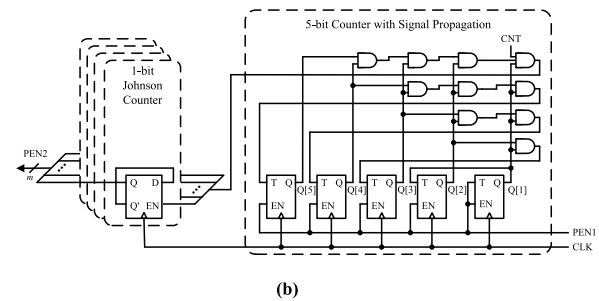
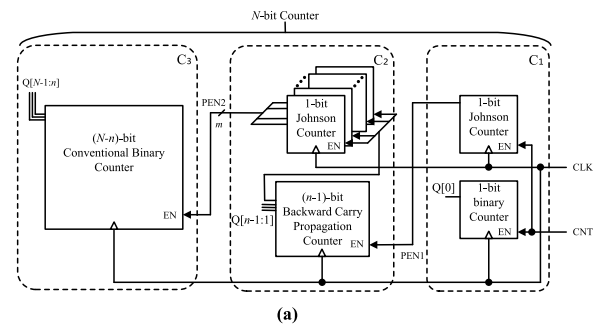


FIGURE 4. Pre-scaled counter with backward signal propagation. (a) Overall architecture, and (b) detailed structure of C_2 .

on T flip-flops that has a relatively large setup time, and thus faces a limitation in achieving faster counting rates.

In this paper, we introduce a new pre-scaled counter that has sub-counters built with the Johnson counter and replaces T flip-flops with D flip-flops to minimize the setup time of the flip-flops. Moreover, we incorporate an early detection method to enhance the counting rate by generating the PEN signal more efficiently. Compared to the state-of-the-art counter, the proposed counter achieves a 33% improvement in speed while maintaining nearly the same hardware complexity.

The rest of this paper is organized as follows: Section II provides a brief introduction to the previous counters. Section III explains in detail the proposed counter architecture. In Section IV, we present the hardware implementation and compare it with the previous counters. Finally, concluding remarks are made in Section V.

II. PREVIOUS COUNTERS

This section describes the previous works related to the proposed binary counter. The conventional synchronous

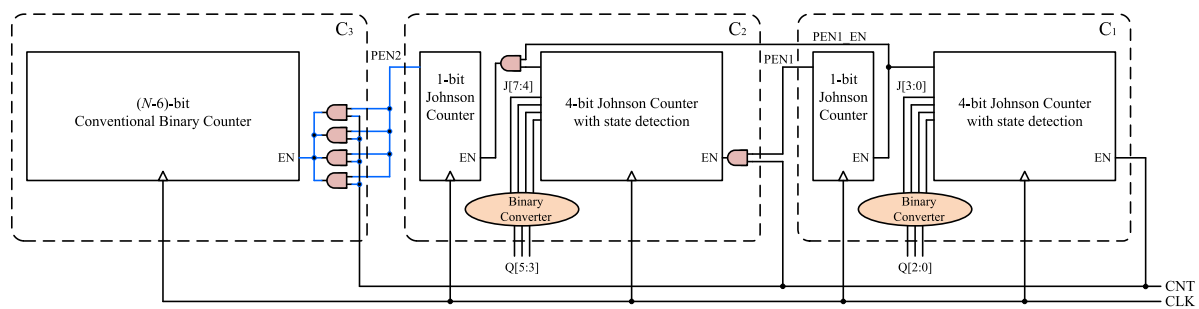


FIGURE 5. Overall architecture of the proposed N -bit counter.

binary counter shown in Fig.1 is typically designed with T flip-flops each of which can be realized with an XOR gate and a D flip-flop. Moreover, there is a chain of AND gates, commonly referred to as a ripple signal chain. The ripple chain propagates the signal from the least significant bit (LSB) to the most significant bit (MSB) in the forward direction, and the output of this chain becomes valid when all the signals are stabilized. However, as the counter size increases, the length of this chain becomes longer, leading to increased delay and decreased counting rate. Various methods have been presented to alleviate the delay of the ripple chain, such as lookahead logic with tree structures, and to enhance flip-flops based on true single-phase clocking [18], [19], [20], [21].

Backward signal propagation [14], depicted in Fig. 2, is a significant concept in the implementation of high-speed synchronous counters. Its core principle lies in leveraging the property of binary numbers, where the more significant bit becomes stable earlier than the less significant bits. Instead of employing a single AND chain, as seen in the conventional binary counter, multiple separate AND chains connected in the reverse order are utilized. This arrangement ensures that the signals stabilized early in the chain are evaluated in advance, making the overall delay of the chains being determined by the last AND gate. However, as the counter size increases, the number of AND chains considered in the backward propagation grows, resulting in a substantial fan-out issue.

A 64-bit binary counter presented in [16] utilizes pre-scaling to achieve a constant delay, as illustrated in Fig. 3. The counter is divided into multiple smaller sub-counters, where the high-order block operates at a lower frequency compared to the low-order block. The low-order block generates a pre-scaled enable (PEN) signal to activate the high-order block, resulting in less frequent increments in the high-order block compared to the low-order block. As a result, the overall counting rate is determined by the delay of the lowest-order block, enabling a nearly constant counting rate for differently sized counters. However, the use of ring counters in generating the PEN signals increases the hardware complexity. Additionally, since a single PEN signal drives

the high-order block, there is a delay issue caused by the large fan-out, which becomes more critical as the counter size increases. Consequently, achieving a constant counting rate in practical implementations becomes challenging for this counter.

A new counter architecture [17], depicted in Fig. 4, increases the counting rate while maintaining a nearly constant counting rate irrespective of counter sizes. This architecture divides an N -bit counter into 1-bit, 5-bit, and $(N-6)$ -bit blocks. To address the hardware complexity associated with the ring counters and the fan-out issue of the PEN signal, multiple 1-bit Johnson counters are utilized. This approach resolves the fan-out issue caused by the upper block and reduces the hardware complexity of the ring counter. The counting rate of the entire counter system is determined by the delay of the 1-bit block, achieving a high counting rate of approximately 2 GHz [17].

However, the counter design is based on T flip-flops which has a relatively high setup time and has a considerable fan-out problem in the 5-bit sub-counter, limiting the potential for the higher counting rates. To mitigate the impact of the setup time and enable faster generation of the PEN signal, a new counting architecture is proposed in this paper. It incorporates 4-bit Johnson counters and an early state detection to generate the PEN signal more efficiently. Unlike the T flip-flop consisting of a D flip-flop and an XOR gate in front of the D input, the 4-bit Johnson counter is built solely with D flip-flops, being effective in reducing the setup time. Additionally, the early state detection technique reduces the critical path, allowing faster generation of the PEN signal.

III. PROPOSED COUNTER ARCHITECTURE

The proposed N -bit counter comprises three sub-counters: C_1 , C_2 , and C_3 , as depicted in Fig. 5. C_1 consists of a 4-bit Johnson counter, a state detection circuit, and a 1-bit Johnson counter that generates a PEN signal denoted as PEN1. As a k -bit Johnson counter has $2k$ states, the 4-bit Johnson counter represents 8 states, which is equivalent to 3 bits in the binary counter. To convert the 4-bit value of the Johnson counter into a binary value, a binary converter is activated when the counter stops counting. C_2 is also constructed with a 4-bit

Johnson counter, a logic circuitry for generating another PEN signal called PEN2. Additionally, there is a binary converter associated with C_2 . Lastly, C_3 is a conventional $(N-6)$ -bit binary counter. Table 1 presents the number of flip-flops required for several combinations of sub-counters differently sized in the proposed method. The combination of C_1 and C_2 representing 3 bits each was selected, as it requires the least number of flip-flops.

TABLE 1. Number of flip-flops for each combination of sub-counters.

sub-counter combination (C_1 bit-width, C_2 bit-width)	(3, 3)	(2, 4)	(1, 5)
Number of flip-flops	4+4	2+8	1+16

A. JOHNSON COUNTER-BASED SUB-COUNTER

The structure of the Johnson counter-based sub-counter, which is the base of C_1 and C_2 , is illustrated in Fig. 6. The middle part is a 4-bit Johnson counter, while the top part is responsible for generating enable signal for next pre-scaled counter, and the bottom part corresponds to a binary converter. The Johnson counter is designed with 4 D flip-flops connected directly without any logic circuits in between, ensuring fast operation. It is worth noting the Q' output of the left-most D flip-flop is directly connected to the input of the right-most D flip-flop. The PEN signal for the next sub-counter should be generated in state 7. Thus, the 1-bit Johnson counter located at the top left of Fig. 6, which generates PEN signal, should be enabled for the last 2 cycles in every 8 cycles. In other words, the 1-bit Johnson counter should be enabled when the 4-bit Johnson counter is in states 6 and 7.

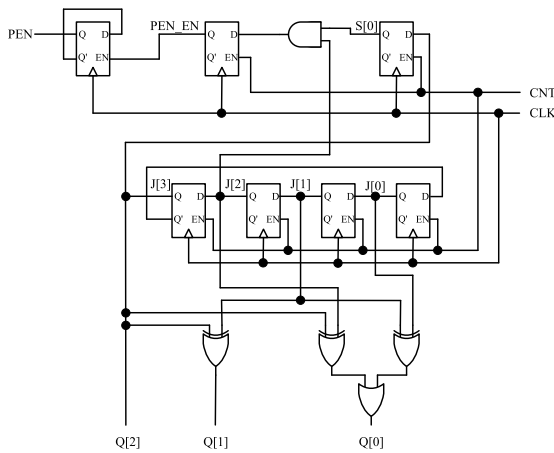


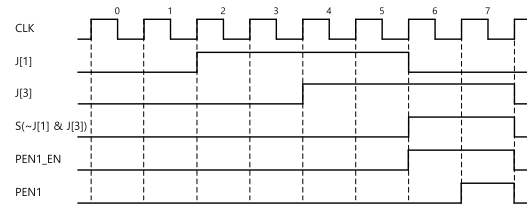
FIGURE 6. Detailed structure of Johnson counter-based sub-counter.

Fig. 7 (a) displays the 4-bit states of the Johnson counter, $J[3:0]$. States 6 and 7 can be identified by examining two bits of the Johnson counter, $J[3]$ and $J[1]$. Therefore, they can be easily detected by ANDing $J[3]$ and $\sim J[1]$, providing a faster mechanism for state detection. Consequently, the PEN generation circuit at the top part of Fig. 6 consists of one

D flip-flop and a 2-input AND gate. The detailed timing of the PEN generation process is depicted in Fig. 7(b).

State	J[3]	J[2]	J[1]	J[0]	PEN_EN
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	1	0
3	0	1	1	1	0
4	1	1	1	1	0
5	1	1	1	0	0
6	1	1	0	0	1
7	1	0	0	0	1

(a)



(b)

FIGURE 7. PEN generation. (a) State detection and (b) timing diagram.

The 4-bit Johnson counter is a good choice, as it is logically equivalent to a 3-bit binary counter. In other words, we can remove the AND chain of the 3-bit binary counter by using one addition flip-flop.

Finally, the 4-bit value $J[3:0]$ of the Johnson counter is converted to a 3-bit binary value $Q[2:0]$ using a few logic gates, as shown at the bottom of Fig. 6. $Q[2]$ is equal to $J[3]$, $Q[1]$ is obtained by XORing $J[3]$ and $J[1]$, and $Q[0]$ by performing an OR operation on the XOR of $J[3]$ and $J[2]$, and the XOR of $J[1]$ and $J[0]$. It is important to note that the binary converter is not a critical path determining the counting rate, as the converted binary value is required only when the counting stops.

B. C_1 AND C_2 SUB-COUNTERS

When considering the connection between C_1 and C_2 , the $PEN1_EN$ signal is utilized to generate the $PEN2_EN$ signal. This implies that two AND gates are connected in series to generate $PEN2_EN$. Since the counting rate is determined by the logic delay between D flip-flops, it is necessary to reduce the delay introduced by the cascaded AND gates. Therefore, the C_1 counter should be modified from basic concept. That means states 5 and 6 are detected instead of states 6 and 7.

The proposed C_1 sub-counter is illustrated in Fig. 8 (a). The detection signal is delayed by one cycle, and the delayed signal is used in C_1 to enable the 1-bit Johnson counter. Hence, the delayed signal serves as $PEN1_EN$. It is important to note that the AND gates in C_1 and C_2 are now separated by a D flip-flop, effectively breaking the cascaded connection. By referring to Fig. 8 (b), it is apparent that states 5 and 6 can be detected by ANDing $J[2]$ and $\sim J[0]$. As the states

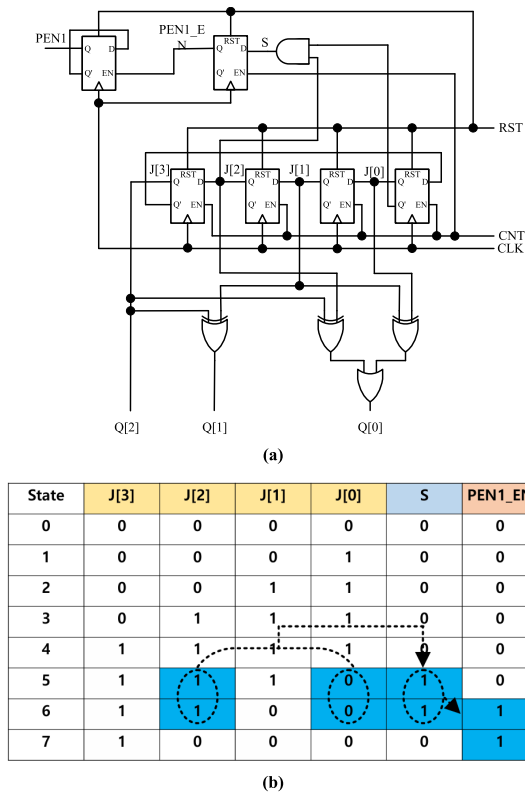


FIGURE 8. Sub-counter C_1 . (a) Detailed structure, and (b) state detection.

are detected one-cycle earlier, the technique is called early state detection.

The C_2 sub-counter shown in Fig. 9(a) comprises a 4-bit Johnson counter, a partial backward propagation circuit for PEN2 generation, and 1-bit Johnson counter that generates PEN2 enabling the C_3 sub-counter. The architecture of C_2 is almost same as Johnson counter-based sub-counter, but the early state detection is excluded. Also, the C_2 counter is activated once every 8 cycles. By examining the state of the Johnson counter in Fig 7(a), we can observe that each bit of the Johnson counter maintains its value for 32 cycles, allowing for a relatively long evaluation period.

The PEN2 signal is generated by activating the 1-bit Johnson counter during the last 2 cycles of every 64 cycles, as depicted in Fig. 9(b). To achieve this, we need to identify state 7 of the 4-bit Johnson counter, which is detected by performing a 3-input AND operation, $J[7] \& \sim J[6] \& \sim J[5]$, as shown in Fig 9(a). State 7 lasts for 8 cycles since the Johnson counter is enabled once every 8 cycles. However, we need to enable the 1-bit Johnson counter for the last two cycle of state 7 in order to generate PEN2. For this purpose, the output of the 3-input AND operation, indicating state 7, is ANDed with PEN1_EN coming from the C_1 sub-counter. It is important to note that the PEN1_EN signal is high during the last 2 cycles of every 8 cycles, resulting in the PEN2_EN signal being high during the last 2 cycles of every 64 cycles. The PEN2_EN signal serves as the enable signal for the 1-bit Johnson counter, ultimately generating the PEN2 signal that

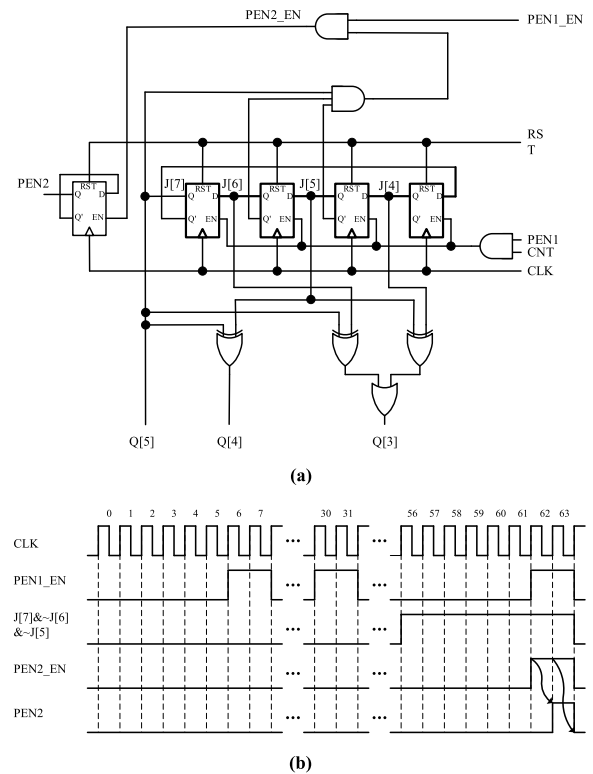


FIGURE 9. Sub-counter C_2 . (a) Detailed structure, and (b) timing diagram.

is activated once every 64 cycles. As the counting process is related with the CNT signal, the PEN2 signal is synchronized with the CNT signal in the C_3 sub-counter. The PEN2 signal generation is depicted in Fig 9(b).

The binary converters at the bottom of Fig. 8(a), and 9(a) generate a 3-bit binary output $Q[5:3]$ and $Q[2:0]$ from the output $J[7:4]$ and $J[3:0]$ of the 4-bit Johnson counters at both C_2 and C_1 .

C. C_3 SUB-COUNTER

The uppermost C_3 sub-counter is depicted in Fig. 10, assuming a 64-bit counter. As a matter of fact, C_3 is a conventional 58-bit binary counter. When combined with the C_1 and C_2 sub-counters, it forms a 64-bit counter. If an N -bit counter is desired, where $N \geq 6$, the counter can be achieved by simply adjusting the size of C_3 to $N-6$ bits. The C_3 sub-counter increments by one every 64 cycles. The bottom part includes some AND gates which make PEN2 signal synchronize with CNT. Thanks to the utilization of multiple AND gates each of which drives up to 12 flip-flops, the fan-out of the PEN2 signal is reduced significantly. This is different from the previous pre-scaled counter [16] equipped with only one ring counter that drives every flip-flop of the upper sub-counter.

D. CRITICAL PATH

Each of the three sub-counters has its own individual critical path. They can be divided into two cases: the first case includes the paths that stabilize internal signals and the

second is related to the generation and usage of the PEN signal. If a path that stabilizes internal signals becomes critical, the critical path of the C_1 counter determines the maximum counting rate of the entire counter for the pre-scaled structure. However, since the PEN signals need to be synchronized with the clock, we need to consider the generation of the PEN signal and its fanout. In order to identify the true critical path and determine the maximum counting rate resulting from it, we will examine potential critical-path candidates for each sub-counter.

In case of C_3 , as it follows a conventional binary counter, the AND chain that should be stabilized before the enabled clock edge becomes the critical path. As the size of the counter increases, the delay in the chain also increases. Since we applied the pre-scaled structure, the C_3 sub-counter will not encounter any issues as long as the delay in the chain remains smaller than 64 times the clock period.

For sub-counter C_2 , the critical-path candidates include the fan-out delay of PEN2, and the generation of PEN2_EN. The PEN2 signal is synchronized with the CNT signal by using 5 AND gates each of which drives up to 12 flip-flops in C_3 . Therefore, the path is determined by the AND gate delay and the fan-out delay caused by 12 flip-flops, excluding the propagation delay of the flip-flop that generates PEN2 and the setup time of the flip-flops in C_3 . The PEN2_EN signal is generated by passing through a 3-input AND gates and a 2-input AND gate. As the Johnson counter is enabled once in 8 cycles, the 3-input AND gate fed from the Johnson counter becomes stable much earlier than the 2-input AND gate that is connected to PEN1_EN. Therefore, the critical path excludes the 3-input AND gate and includes the 1-bit Johnson counter and the 2-input AND gate. As mentioned earlier, the binary converters in C_1 and C_2 are not critical at all, because the converters are needed only when the counting stops.

In case of C_1 , there are two potential candidates for the critical path. The first candidate is the state detection circuit depicted in Fig. 8, which detects states 5 and 6 using a 2-input AND gate. The path is formed with a 2-input AND gate, excluding flip-flops. The path formed from the D flip-flop to the 1-bit Johnson counter generating PEN1 is not critical as there is no logic gate on it. The second candidate is related to the propagation of the PEN1 signal, which encompasses the AND gate needed to synchronize with CNT, and four D flip-flops that are enabled by PEN1 in the C_2 sub-counter.

The critical-path delays of the proposed 64-bit counter are summarized in Table 2, which includes not only gate delays between D flip-flops but also the setup time and propagation delay of the involving flip-flops. For C_2 , there are two critical path candidates. The first candidate generates PEN2_EN based on partial backward propagation. This path is composed of the 2-input AND gate and connected to a D flip-flop, resulting in 0.23ns. The second candidate is the propagation of PEN2. The PEN2 signal is synchronized with CNT by using 5 AND gates in C_3 and connected to enable 12 flip-flops in C_3 , resulting in 0.27ns.

TABLE 2. Critical-path delay of sub-counters.

sub-counter	paths	delay [ns]
C_1	PEN1 signal path	0.25
	PEN1_EN generating logic	0.23
	Binary converting	0.19
C_2	PEN2 signal path	0.27
	PEN2_EN generating logic	0.23
	Binary converting	0.19
C_3	AND chain	1.53

In C_1 , there are two critical-path candidates. The first is related to the propagation of PEN1 that enables C_2 , and the second is the early state detection required to generate PEN1_EN. As summarized in Table 2, their delays are 0.25ns and 0.23ns, respectively. The binary converters in C_1 and C_2 are not critical as described earlier, and the delay is even smaller than the critical-path candidates.

The C_3 counter has a long AND chain as shown in Fig. 10. As C_3 is enabled once every 64 cycles, however, it is sufficient to stabilize the chain for 64 cycles. The delay of the chain is 1.53ns, which is much shorter than 64 cycles with a counting rate of 0.27ns. As a result, the delay of the chain does not matter at all, which is the property enabling the pre-scaled counter structure. In Table 2, the critical path is formed in propagating PEN2, which is depicted with blue lines in Fig. 5. The proposed counter achieves a counting rate of 3.7GHz, which is much faster than the previous counters.

IV. IMPLEMENTATION AND COMPARISON

In this section, we evaluate the proposed counter and compare its performance to those of the conventional binary counter, the LFSR counter [13], and the pre-scaled counter with backward signal propagation [17]. The performances were analyzed in terms of the maximum counting rate, hardware complexity (area), and power consumption while varying the counter size from 8 to 64 bits. The maximum counting rate is determined by considering the critical-path delay and the setup time and propagation delay of flip-flops. All the counters were implemented in a 28nm CMOS technology and synthesized with standard cells using Synopsys Design Compiler, and the performances of the counters were measured under a uniform environment at an operating voltage of 1V and a temperature of 25°C.

The maximum counting rates simulated under the conditions of 1V and 25°C are depicted in Fig. 11 for various counter sizes. For the conventional binary counter, the length of the AND chain increases as the counter size increases, resulting in an increase in the critical-path delay and a decrease in the counting rate. On the other hand, the pre-scaled counter with backward propagation maintains an almost constant counting rate regardless of the counter size, as reported in [17]. The previous pre-scaled counter [17] employs T flip-flops in the 1-bit and 5-bit sub-counters resulting a large setup time compared to that of

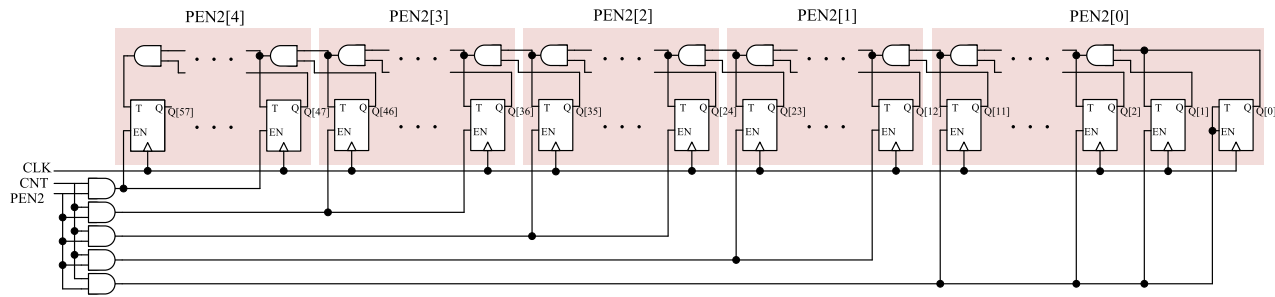


FIGURE 10. Sub-counter C_3 .

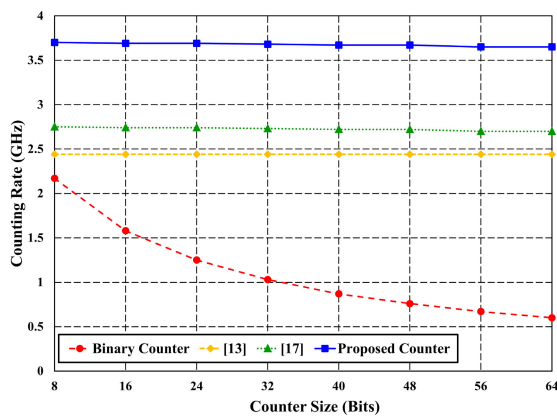


FIGURE 11. Maximum counting rate.

D flip-flops. For a 64-bit counter, the maximum counting rate of the previous pre-scaled counter [17] is 2.7 GHz. In the proposed counter, 4-bit Johnson counters are used as they utilize only D flip-flops. Since a D flip-flop has the lower setup time than a T flip-flop, the proposed counter can operate much faster. In addition, it has a pre-scaled structure, resulting in an almost constant counting rate independent of the counter size.

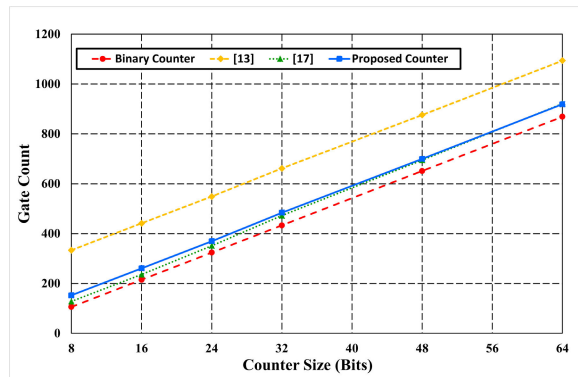


FIGURE 12. Hardware complexity in equivalent gate count.

The hardware complexity is compared in terms of the number of equivalent gates (EGs) required to implement

TABLE 3. Hardware complexities of four counters.

Counter Size (Bits)	Counter Type	Combinational Equivalent Gates	D-FFs
8	Binary	40	8
	[13]	218	14
	[17]	45.5	10
	Proposed	45.5	13
16	Binary	83	16
	[13]	260	22
	[17]	87.5	18
	Proposed	87.5	21
24	Binary	126	24
	[13]	301.25	30
	[17]	128.75	27
	Proposed	130.25	29
32	Binary	169	32
	[13]	348	38
	[17]	175.5	36
	Proposed	178.5	37
48	Binary	255	48
	[13]	430.25	54
	[17]	257.75	53
	Proposed	262.25	53
64	Binary	341	64
	[13]	516.25	70
	[17]	343.75	70
	Proposed	349.75	69

combinational logic circuits and flip-flops. An EG corresponds to a 2-input NAND gate, and the complexity is summarized and compared with other counters in Table 2. According to the standard cell library used in the implementation, an AND gate, an XOR gate, an inverter, and a D flip-flop correspond to 1.25, 2.25, 0.75, and 8.25 EGs, respectively. The total EG counts, obtained by summing combinational circuits and flip-flops, are depicted in Fig. 12. The proposed counter requires an additional D flip-flop for state detection in C_1 , partial backward propagation, and AND gates for CNT synchronization, while the previous pre-scaled counter [17] utilizes T flip-flops, a 5-bit binary counter with backward signal propagation, and multiple Johnson counters. The higher EG count leads to a larger area, so the proposed counter is slightly larger than the previous pre-scaled counter if the counter size is small. However, the increase is smaller and becomes negative when the counter size is large.

TABLE 4. Synthesis results for various counters.

Counter Size (Bits)	Counter Type	Technology	Delay [ns] (Normalized)	Area [μm^2] (Normalized)	Area-Delay Product [$\mu\text{m}^2 \cdot \text{ns}$] (Normalized)	Total Power [mW] (Normalized)	Gate Power [mW] (Normalized)	Flip-flop Power [mW] (Normalized)
8	Binary	28nm	0.41 (1.52)	49.61 (0.80)	20.34 (1.21)	0.1375 (5.37)	0.0166 (6.15)	0.1209 (5.30)
	[13]	28nm	0.41 (1.52)	144.38 (2.32)	59.20 (3.52)	0.0531 (2.08)	0.0179 (6.59)	0.0352 (1.54)
	[17]	28nm	0.36 (1.33)	67.74 (1.09)	24.39 (1.45)	0.0289 (1.13)	0.0033 (1.21)	0.0257 (1.12)
	Proposed	28nm	0.27 (1.00)	62.36 (1.00)	16.84 (1.00)	0.0256 (1.00)	0.0027 (1.00)	0.0228 (1.00)
16	Binary	28nm	0.59 (2.19)	100.62 (0.83)	59.00 (1.81)	0.1644 (6.27)	0.0274 (9.79)	0.1370 (5.88)
	[13]	28nm	0.41 (1.52)	194.92 (1.62)	79.92 (2.45)	0.0537 (2.03)	0.0180 (6.34)	0.0357 (1.53)
	[17]	28nm	0.36 (1.33)	125.89 (1.04)	45.32 (1.39)	0.0295 (1.13)	0.0034 (1.20)	0.0261 (1.12)
	Proposed	28nm	0.27 (1.00)	120.63 (1.00)	32.57 (1.00)	0.0262 (1.00)	0.0028 (1.00)	0.0233 (1.00)
24	Binary	28nm	0.77 (2.85)	151.63 (0.89)	116.76 (2.53)	0.1974 (7.37)	0.0274 (9.13)	0.1618 (6.80)
	[13]	28nm	0.41 (1.52)	245.11 (1.43)	100.50 (2.18)	0.0544 (2.03)	0.0182 (6.09)	0.0362 (1.52)
	[17]	28nm	0.36 (1.33)	176.09 (1.03)	63.39 (1.37)	0.0302 (1.13)	0.0035 (1.19)	0.0266 (1.12)
	Proposed	28nm	0.27 (1.00)	170.82 (1.00)	46.12 (1.00)	0.0268 (1.00)	0.0030 (1.00)	0.0238 (1.00)
32	Binary	28nm	0.97 (3.59)	202.64 (0.91)	196.56 (3.26)	0.2255 (8.23)	0.0438 (14.12)	0.1817 (7.48)
	[13]	28nm	0.41 (1.52)	297.88 (1.33)	122.13 (2.02)	0.0550 (2.01)	0.0183 (5.90)	0.0357 (1.51)
	[17]	28nm	0.36 (1.33)	228.85 (1.02)	82.39 (1.36)	0.0308 (1.12)	0.0037 (1.18)	0.0271 (1.12)
	Proposed	28nm	0.27 (1.00)	223.59 (1.00)	60.37 (1.00)	0.0274 (1.00)	0.0031 (1.00)	0.0243 (1.00)
48	Binary	28nm	1.32 (4.89)	304.67 (0.94)	402.16 (4.60)	0.2831 (9.70)	0.0593 (16.94)	0.2238 (8.71)
	[13]	28nm	0.41 (1.52)	397.10 (1.23)	162.81 (1.86)	0.0567 (1.95)	0.0187 (5.36)	0.0381 (1.48)
	[17]	28nm	0.36 (1.33)	329.12 (1.02)	118.48 (1.35)	0.0325 (1.12)	0.0041 (1.16)	0.0285 (1.11)
	Proposed	28nm	0.27 (1.00)	323.86 (1.00)	87.44 (1.00)	0.0292 (1.00)	0.0035 (1.00)	0.0257 (1.00)
64	Binary	28nm	1.67 (6.19)	406.70 (0.95)	679.19 (5.86)	0.3171 (10.64)	0.0642 (17.35)	0.2529 (9.69)
	[13]	28nm	0.41 (1.52)	503.57 (1.17)	206.46 (1.78)	0.0574 (1.93)	0.0189 (5.08)	0.0384 (1.47)
	[17]	28nm	0.36 (1.33)	432.20 (1.01)	155.59 (1.34)	0.0332 (1.11)	0.0043 (1.15)	0.0289 (1.11)
	Proposed	28nm	0.27 (1.00)	429.27 (1.00)	115.90 (1.00)	0.0298 (1.00)	0.0037 (1.00)	0.0261 (1.00)

The synthesis results of the proposed counter are compared with those of previous counters, as summarized in Table 4. To ensure a fair comparison, all counters were simulated at the maximum counting rate of the 64-bit conventional binary counter, which is 600 MHz. The proposed counter achieves the lowest delay while maintaining a comparable area, leading to superior performance in terms of the area-delay product. It was observed that the power consumed by flip-flops is significantly higher than that dissipated in the gates. As a result, the proposed counter exhibits the lowest overall power consumption among the evaluated designs.

The proposed 64-bit counter exhibits an approximately 14% and 1% reduction in area compared to the previous LFSR and pre-scaled counters, while showing a 5% increase in area relative to the binary counter. Additionally, it achieves a 48%, 25%, and 91% reduction in power consumption compared to the LFSR counter, the pre-scaled counter, and the binary counter, respectively. Most importantly, the proposed counter operates about 33% faster than the previous state-of-the-art pre-scaled counter [17].

V. CONCLUSION

This paper has proposed a high-speed synchronous binary counter architecture capable of achieving a constant counting rate regardless of the counter size. The proposed architecture utilizes 4-bit Johnson counters constructed exclusively with D flip-flops, effectively reducing the overall setup time. Furthermore, the early state detection technique was incorporated to facilitate significantly faster generation of the PEN signal. Although the proposed counter slightly increases the

area and power consumption compared to the previous pre-scaled counter, these increments are negligible in practical applications. The proposed counter demonstrates a counting rate of 3.7 GHz in a 28 nm CMOS process, independent of the counter size, achieving approximately 33% higher speed than the state-of-the-art pre-scaled counter.

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